

ASHGABAT, Turkmenistan

WMO: 388800

Lat: 37.987N

Lon: 58.361E

Elev: 692

StdP: 14.33

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.0	23.4	10.0	9.4	27.8	14.4	11.7	35.1	20.4	44.9	17.3	44.4	5.5	100	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	105.7	67.8	103.0	67.1	100.7	66.7	74.0	95.0	72.4	93.5	70.8	92.3	9.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	100.4	86.9	64.6	93.5	85.6	62.6	87.2	83.6	38.1	94.7	36.5	93.7	35.0	92.2	92.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.2	15.8	DB	14.2	110.0	7.2	2.6	9.0	111.8	4.8	113.3	0.7	114.8	-4.6	116.6	(4)
(5)			WB	13.3	77.3	7.2	3.6	8.1	79.9	3.9	82.0	-0.2	84.0	-5.4	86.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	39.5	42.5	52.6	63.8	75.5	84.6	88.6	86.1	76.0	63.7	50.5	41.7		
	DBStd	18.97	8.27	9.25	8.64	8.67	6.35	5.14	4.38	4.87	6.29	8.02	8.53	8.14		
	HDD50	1028	337	235	70	7	0	0	0	0	0	6	94	279		
	HDD65	3229	789	629	396	123	6	0	0	0	4	119	441	722		
	CDD50	6100	13	26	150	420	792	1038	1197	1120	781	432	108	23		
	CDD65	2826	0	0	11	86	333	588	732	655	335	80	5	1		
	CDH74	39704	0	9	198	939	3809	8553	11469	9778	3998	875	68	8		
	CDH80	23146	0	2	66	366	1815	5106	7402	6110	1960	300	18	1		
(14) Wind	WSAvg	6.9	5.7	6.4	7.1	7.6	7.8	8.3	8.2	7.4	6.8	6.1	5.5	5.6		
(15) Precipitation	PrecAvg	9.2	0.9	1.2	1.6	1.6	1.1	0.3	0.1	0.0	0.1	0.5	0.8	0.8		
	PrecMax	15.7	2.8	3.3	3.9	4.6	3.3	1.7	1.1	0.4	1.2	1.6	2.8	2.2		
	PrecMin	5.6	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.3	0.7	0.8	0.7	1.2	0.9	0.4	0.2	0.1	0.2	0.5	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	74.7	87.8	94.9	100.6	107.2	109.5	108.0	100.8	94.6	81.8	73.7		
		MCWB	51.0	53.2	59.3	63.5	66.2	66.4	69.7	68.1	64.6	62.6	58.5	53.3		
	2%	DB	62.8	67.7	79.1	89.3	96.5	103.7	106.0	104.4	96.8	87.4	73.6	65.9		
		MCWB	48.3	50.3	56.7	62.1	65.0	66.8	68.5	67.0	64.1	61.4	55.5	50.0		
	5%	DB	56.1	61.2	73.3	84.0	93.0	100.7	103.7	102.1	93.5	82.6	68.3	58.8		
		MCWB	46.0	48.4	55.2	60.9	64.0	65.8	67.8	66.6	64.2	59.9	54.0	47.9		
	10%	DB	51.2	56.1	67.6	78.9	89.6	98.4	100.8	99.3	90.0	78.7	64.1	53.5		
		MCWB	44.0	46.7	53.7	59.3	63.3	65.6	67.8	66.3	63.2	58.2	52.4	45.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	55.1	61.4	67.5	71.4	72.7	76.4	75.6	71.7	66.5	61.0	54.2		
		MCDB	64.9	68.9	80.4	85.2	87.7	94.8	97.4	96.8	91.5	82.5	74.1	68.8		
	2%	WB	49.1	51.8	58.4	64.7	68.0	70.8	74.5	73.7	69.3	63.9	58.1	51.0		
		MCDB	60.4	63.7	74.6	82.1	87.4	94.0	96.1	94.3	87.7	80.9	69.3	62.4		
	5%	WB	46.7	49.6	56.2	62.8	66.6	69.5	73.0	71.9	67.5	62.1	55.8	48.6		
		MCDB	55.3	59.2	70.1	78.7	85.7	92.7	94.4	92.3	86.0	77.8	65.2	57.1		
	10%	WB	44.4	47.4	54.1	60.9	65.4	68.3	71.4	69.9	65.7	60.1	53.5	46.4		
		MCDB	50.4	54.9	65.8	75.5	83.6	90.7	93.3	90.7	84.6	75.2	62.6	52.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.3	18.9	20.2	22.0	23.2	23.0	24.6	24.6	22.7	18.6	15.8		
		MCDBR	25.8	27.9	30.2	28.5	27.6	27.9	26.5	27.3	28.3	27.9	26.5	26.3		
		MCWBR	14.3	14.7	12.9	10.4	9.5	9.7	9.4	10.3	10.8	11.5	12.7	14.5		
		MCDBR	22.8	24.1	26.2	23.4	22.7	21.9	20.1	21.0	21.5	23.7	21.9	22.6		
	5% WB	MCWBR	13.3	13.4	12.4	10.0	8.9	8.7	9.4	10.2	10.4	10.8	11.4	13.2		
(40) Clear-Sky Solar Irradiance	taub		0.419	0.478	0.571	0.632	0.634	0.624	0.630	0.570	0.530	0.548	0.481	0.424		
	taud		2.006	1.860	1.690	1.622	1.649	1.669	1.662	1.762	1.818	1.774	1.893	1.993		
	Ebn at Noon		226	228	219	216	219	221	219	228	227	203	201	213		
	Edh at Noon		41	54	71	80	79	78	78	69	61	58	46	39		
(44) All-Sky Solar Radiation	RadAvg		694	952	1313	1729	2129	2498	2454	2285	1871	1302	855	630		
	RadStd		72	109	102	173	105	75	73	93	55	77	72	82		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.74	N/A	N/A	+1.40	N/A	N/A	N/A	N/A	+202	+182
(47) Regional (4 neighbors)		N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A	N/A	+148

Nomenclature: See separate page